Microwave-assisted CO oxidation over a perovskite oxide catalyst - a surface spectroscopic study

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In this study, we investigated the microwave-assisted (MW) CO oxidation over a (La,Sr)$CoO_{3-\delta}$ (LSC) perovskite oxide catalyst under dry and humidified conditions. The reaction serves a model for the energy efficient aftertreatment of exhaust gas components and the use of MW as energy source can offer multiple advantages over traditional thermocatalysis (TC), as the nature of the MW-solid interaction offers quick, adaptive and energy efficient heating as well as improved yield and lower light-off temperatures. During the catalytic tests we found that the catalytic performance of this reaction was significantly improved under MW compared to TC and this effect could be related to an initial activating MW pretreatment. Our study indicates that a surface reaction can be driven by the use of MW-radiation in a similar magnitude that can be achieved by thermal activation at significantly higher temperatures. The underlying physicochemical reason was investigated by X-Ray absorption (XAS) and X-Ray photoelectron spectroscopy (XPS) and we found evidence for a significant structural, chemical and electronic reorganization of the oxide surface, possibly consistent with the occurrence of locally overheated surfaces or "hotspots" during MW-exposure. The good catalytic performance, quick response to MW-heating and long-term stability of the catalyst all indicate the promising potential of a MW-based process for the energy efficient exhaust aftertreatment using noble-metal-free oxide catalysts.